PUNE INSTITUTE OF COMPUTER TECHNOLOGY

DHANKAWADI, PUNE – 43.

LIST OF LAB ASSIGNMENTS

ACADEMIC YEAR: 2026-2027

DEPT: ARTIFICIAL INTELLIGENCE & DATA SCIENCE **Date: 01/07/2026**
CLASS: B.E. **SEMESTER: VII**
SUBJECT: Computer Laboratory-II

Sr. No.	PROBLEM STATEMENT
Group A :QAI(Quantum Artificial Intelligence)	
1.	Implementations of 16 Qubit Random Number Generator
2.	Tackle Noise with Error Correction
3.	Implement Tarrataca's quantum production system with the 3-puzzle problem
4.	Implement Quantum Teleportation algorithm in Python
5.	Implementing a 5 qubit Quantum Fourier Transform
Group B: EAC(Enterprise Architecture &Components)	
6.	Write a short report on planning, securing, and governing the enterprise architecture.
7.	Sketch enterprise architecture with emerging technologies such as cloud / IoT / AI / Blockchain.
8.	Design and implement an enterprise architecture framework for a hypothetical organization, considering the key components such as business architecture and technology architecture.
9.	Design an enterprise information architecture that includes a detailed component model and operational model.

10.	Generate a comprehensive report on open source Enterprise Architecture Tools - LeanIX Enterprise Architecture Management, ADOIT EA Suite, UPMX, Avolution ABACUS
Group C: UI/UX Design	
13	Design user persona for the users of selected product / system. II. How To Create A User Persona (Video Guide) - YouTube III. How to Create A User Persona in 2022 [FULL GUIDE] - YouTube
14	Online Learning Platform: Design a wireframe for an online learning platform that includes course listings, video lectures, quizzes, and progress tracking. I. E-learning Website Design in Figma - YouTube
15	Designing a Social Fitness App: Create wireframes and a prototype for a social fitness app that allows users to track workouts, connect with friends, and share progress. Design the user interface for logging exercises, setting goals, and incorporating social features. I. Fitness App Design In Figma || Figma Tutorial || Design & Prototyping - YouTube
16	Use Figma tool to Design a user interface for a recipe finder application, allowing users to search for recipes based on ingredients, categories, and dietary restrictions. Include features like recipe details, cooking instructions, and saving favorites. I. Create a Food & Drink Recipe app with reviews from Figma no code - YouTube
17	Usability Testing Simulation: Develop a high-fidelity interactive prototype using any UI/UX tool. Prepare a usability testing plan, recruit participants, and simulate usability testing sessions. Analyze the feedback and iterate on the design based on the insights gathered during the testing. Usability Testing in UX Design Thinking Process - YouTube

Group D: IR(Information Retrival)	
18	Write a program for pre-processing of a text document such as stop word removal, stemming.
19	Implement a program for retrieval of documents using inverted files.
20	Implement e-mail spam filtering using text classification algorithm with appropriate dataset.
21	Implement Agglomerative hierarchical clustering algorithm using appropriate dataset.
22	Implement Page Rank Algorithm. (Use python or beautiful soup for implementation).

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